CS728: Assignment 1

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February 14, 2026

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Chapter 1

Task 1

The following table shows the final parameters used for the Glove training:

Hyperparameter	Value
Window Size (w)	10
Learning Rate (η)	0.05
Epochs	25
$x_{\max}$	100
α	0.75

Table 1.1: Selected GloVe hyperparameters

Nearest Neighbours for this task have been mentioned with results of Task 2.

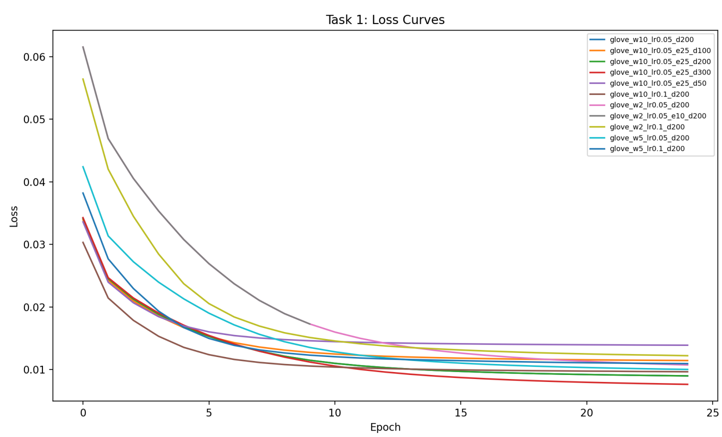

Figure 1.1: Loss curves for different embedding dimensions

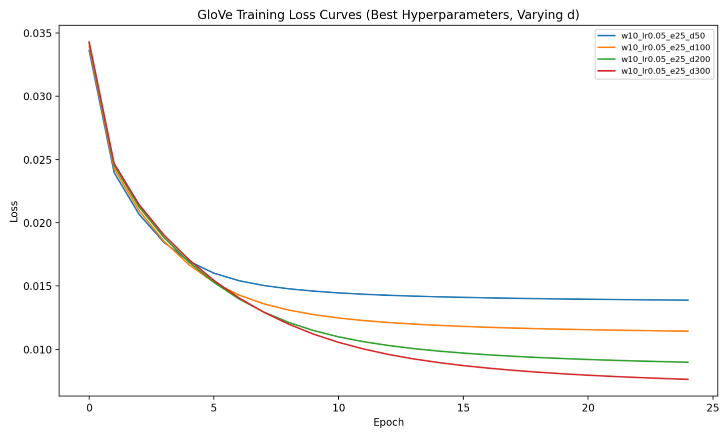

Figure 1.2: Loss curves for best hyperparameters across embedding dimensions

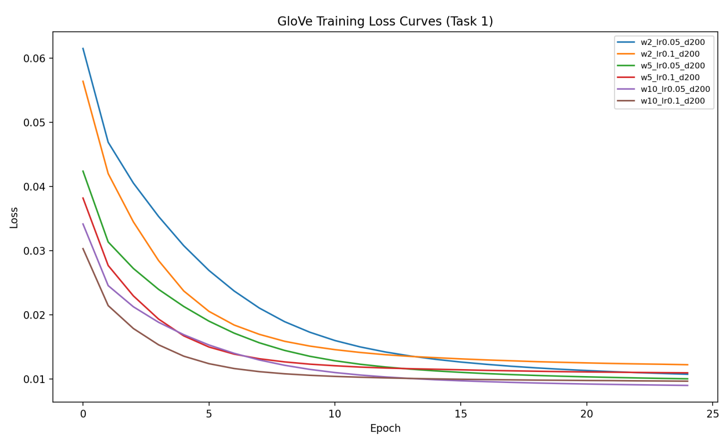

Figure 1.3: Loss curves for embedding dimension 200

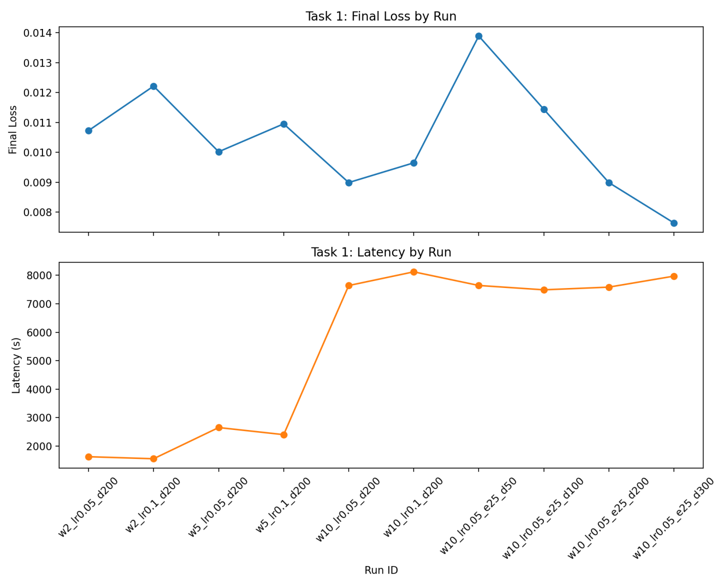

Figure 1.4: Final loss and Latency for different hyperparameters

Chapter 2

Task 2

Model	Word	Top-5 Neighbours
GloVe (d=200)	data	showed, statistics, tracking, according, information
	cities	towns, counties, across, areas, twin
	temperature	temperatures, maximum, levels, rainfall, room
SVD (d=50)	data	consumers, risk, tracking, technology, exposure
	cities	urban, influx, airports, communities, places
	temperature	heating, severe, fog, checking, causes
SVD (d=100)	data	automated, rapid, tool, specific, tracking
	cities	urban, transit, places, tourism, suburbs
	temperature	skin, heating, pump, cold, dry
SVD (d=200)	data	infrastructure, authentication, formulated, users, est
	cities	urban, mayor, city, transit, vast
	temperature	cold, slow, evaporate, brush, water
SVD (d=300)	data	mainframe, users, collected, infrastructure, breach
	cities	urban, city, mayor, vast, places
	temperature	cold, temperatures, evaporate, surface, fluids

Table 2.1: Top-5 nearest neighbours for selected words in GloVe and SVD models

Chapter 3

Task 3

Chapter 4

Task 4

We used a simple MLP architecture with a single hidden layer with 128 neurons followed by a ReLU and a dropout layer with a dropout rate of 0.2. The final output layer has 9 neurons corresponding to the 9 classes in the dataset. For OOV words we use the mean of the embeddings of the words in the training set as the embedding for the OOV word as it acts like a neutral embedding and does not bias the model towards any particular class. We trained the model for 10 epochs with a batch size of 1024 and a learning rate of 0.001 using the Adam optimizer. The tables below show the results of our model on the test set.

Dimension	Accuracy	Weighted F1	Macro F1
50	0.8309	0.7619	0.2179
100	0.8314	0.7643	0.2198
200	0.8314	0.7653	0.2224
300	0.8319	0.7658	0.2241

Table 4.1: Test performance of GloVe-MLP across embedding dimensions

Dimension	Accuracy	Weighted F1	Macro F1
50	0.8274	0.7523	0.2043
100	0.8301	0.7589	0.2091
200	0.8309	0.7609	0.2126
300	0.8305	0.7607	0.2118

Table 4.2: Test performance of SVD-MLP across embedding dimensions

From this we see that the best performance is achieved with the GloVe embeddings with a dimension of 300 and the SVD embeddings with a dimension of 200. **TODO Add reason why CRF performs better**

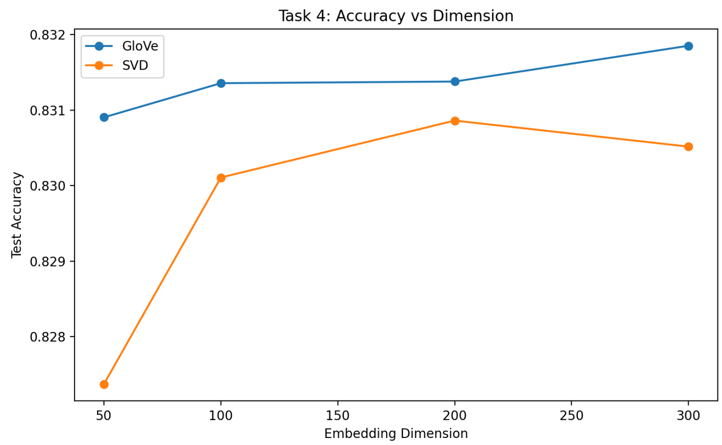

Figure 4.1: Test accuracy across embedding dimensions

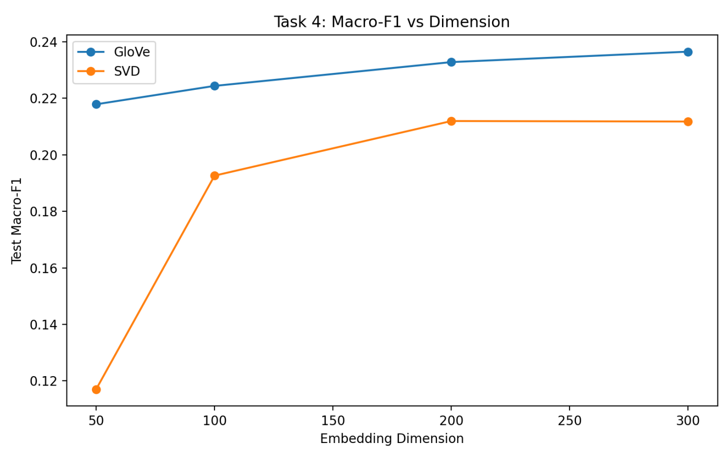

Figure 4.2: Test F1 score across embedding dimensions

Chapter 5

Task 5

Model	Word	Top-5 Neighbours
Raw SVD (d=200)	data	infrastructure, authentication, formulated, users, est
	temperature	cold, slow, evaporate, brush, water
	cities	urban, mayor, city, transit, vast
	apple	Not in vocabulary
	fast	enough, hard, keep, make, stopping
TFIDF SVD (d=200)	data	formulated, infrastructures, reads, est, mentioning
	temperature	cold, evaporate, temperatures, climatologist, gases
	cities	urban, inexorable, mayor, city, vast
	apple	Not in vocabulary
	fast	enough, hard, quick, make, up

Table 5.1: Comparison of Top-5 nearest neighbours for selected words in raw-SVD and TFIDF-SVD models